

JIACHENG LIANG

Curriculum Vitae

CONTACT

✉ jliang66@jh.edu

Homepage: 552jc.github.io/ljc552.github.io

Department of Mathematics, Krieger 211
Johns Hopkins University

EDUCATION

Johns Hopkins University

2024/08 – now

Ph.D. in Mathematics

Advisor: David Gepner

Southern University of Science and Technology, Shenzhen, China

2022/09 – 2024/06

Master in Mathematics

Advisor: Yifei Zhu

Sichuan University, Chengdu, Sichuan, China

2018/09 – 2022/06

B.S. in Physics

VISITS

Max Planck Institute for Mathematics, Bonn, Germany

2025/01/20 – 2025/05/01

Visiting scholar

RESEARCH INTERESTS

My research interests lie broadly in stable homotopy theory, higher algebra, higher category theory, algebraic geometry, and algebraic K -theory. I am particularly interested in localization and finiteness phenomena in large ∞ -categories, including smashing localizations, dualizable categories, and their applications to chromatic homotopy theory and analytic geometry.

PREPRINTS AND PUBLICATIONS

Generalized Telescope Conjecture (Preprint, 2026)

Dualizable Additive Categories (joint with Ishan Levy and Vladimir Sosnilo; Preprint, 2026)

Smashing, Balmer, Zariski Spectra: An Ideal Approach (joint with Changan Zou; Preprint, 2026)

Higher Algebra in t -Structured Tensor Triangulated ∞ -Categories (to appear in *Selecta Mathematica*)

SELECTED TALKS

Dualizable Additive Categories

October 13, 2026

Invited talk, Algebraic Topology Seminar, University of Chicago

Generalized Telescope Conjecture

October 12, 2026

Invited talk, Topology Seminar, Northwestern University

Higher Algebra in t -Structured Tensor Triangulated ∞ -Categories

April 14, 2025

Topology Seminar, Max Planck Institute for Mathematics, Bonn

∞ -topoi and parametrized homotopy theory

June 17, 2024

Graduate Topology Seminar at SUSTech

Picard ∞ -groupoids, Picard groups of E_∞ -rings and generalized Thom spectra April 23, 2024

Graduate Topology Seminar at SUSTech

Barr–Beck theorem, Morita theory and Brauer groups in ∞ -categories March 19, 2024

Graduate Topology Seminar at SUSTech

An overview of ∞ -categories and higher algebra December 26, 2023

Graduate Topology Seminar at SUSTech

The σ -orientation and its AHR E_∞ -refinement $MString \rightarrow tmf$ August 17, 2023

IWoAT Summer School 2023: Operads, Spectra, and Multiplicative Structures

BIMSA, Beijing, China

SEMINAR ORGANIZATION

- **Reading Seminar on Chromatic Homotopy**, Johns Hopkins University Fall 2026
- **Reading Seminar on Prismatic Cohomology**, Johns Hopkins University Fall 2025
- **Reading Seminar on Dualizable Categories**, Johns Hopkins University Fall 2024
- **Graduate Topology Seminar**, Southern University of Science and Technology 2022–2024
- **Undergraduate Topology Seminar**, Sichuan University 2019–2022

TEACHING

- Teaching Assistant for Ordinary Differential Equations, Johns Hopkins University, Fall 2026.
- Teaching Assistant for Abstract Algebra, Johns Hopkins University, Spring 2026.
- Teaching Assistant for Linear Algebra, Johns Hopkins University, Fall 2025.
- Teaching Assistant for Ordinary Differential Equations, Southern University of Science and Technology, Spring 2023.

MISCELLANEOUS

- **Blog** about algebraic topology: www.zhihu.com/people/liang-jia-cheng-60-11/posts